TIANRU ZHANG

Postdoc at Uppsala University

I am a postdoc in the division of system and control, at the IT department of Uppsala University. My research topics include Multimodal LLM, Federated earning and AI-enhanced Data Management. With a strong background in Deep Learning, Reinforcement Learning, Cloud Computing, and Data Management, I have contributed to cutting-edge researches and published extensively in these areas. My professional experiences also include impactful projects in data analysis pipeline, predictive model, and application in various domains.

CONTACT

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https://github.com/JSFRi

Tianru Zhang

SKILLS

- Data Management
- Machine Learning
- Cloud Computing
- Python, C++, MATLAB
- Academic Teaching

LANGUAGES

- English (Fluent)
- Chinese (Native)
- French (Conversational)
- Swedish (Entry)

EDUCATION

- **2020~2025**
  Uppsala Univ., Sweden
  Ph.D., Scientific Computing
- **2017~2018**
  ENSAI, CREST, France
  M.Sc., Data Science
- **2013~2017**
  USTC, China
  B.Sc., Mathematics

PUBLICATIONS

- **Zhang, T.**, Hellander, A., and Toor, S. "Efficient hierarchical storage management empowered by reinforcement learning." *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2022
- **Zhang, T.**, *et al*. "Data management of scientific applications in a reinforcement learning-based hierarchical storage system." *Expert Systems with Applications*, 2023
- **Zhang, T.** "Autonomous hierarchical storage management via reinforcement learning." *Proceedings of the VLDB Endowment*, 2024
- **Zhang, T.**, *et al*. "ReStore: A Reinforcement Learning Approach for Data Migration in Multi-Tiered Storage." *ACM SIGMOD/PODS International Conference on Management of Data*, 2026
- **Zhang, T.**, *et al*. "InfoHier: Hierarchical Information Extraction via Encoding and Embedding." *Preprint https://arxiv.org/abs/2501.08717*
- Ju, L., **Zhang, T.**, *et al*. "Accelerating fair federated learning: Adaptive federated adam." *IEEE Transactions on Machine Learning in Communications and Networking*, 2024
- Li, S., Ngai, E., Ye, F., Ju, L., **Zhang, T.**, and Voigt, T. "Blades: A unified benchmark suite for byzantine attacks and defenses in federated learning." *IEEE/ACM Ninth International Conference on Internet-of-Things Design and Implementation (IoTDI)*, 2024

WORK EXPERIENCE

- **Fujitsu R&D center | 2019~2020**
  Beijing, China
  - Multi-factor classification of personal loan clients
  - Radial Basis Function Neural Network for DeepTensor©
- **IRT SystemX | 2018~2019**
  Paris, France
  - Decentralized Machine Learning for urban mobility
  - Trace prediction in public transport using HMM